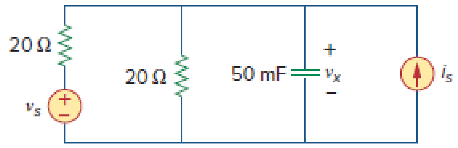
Chapter 10, Problem 44P

Problem

Use the superposition principle to obtain *vx* in the circuit of Fig. 10.89. Let *vs* = 50 sin 2*t* V and *is* = 12 cos(6*t* + 10°) A.

Figure 10.89:



Step-by-step solution

Step 1 of 6

Refer to Figure 10.89 in the textbook.

The circuit operates at two different frequencies; one way to obtain a solution is to use superposition theorem, which breaks the problem into single – frequency problems.

So let,

$v_x = v_1 + v_2$ ...... (1)

Where,

v_1 is due to the $v_s = 50\sin(2t)\,\text{V}$ voltage source,

v_2 is due to the $i_s = 12\cos(6t + 10^0)\,\text{A}$ current source.

Step 2 of 6

To find v_1, consider the voltage source $v_s = 50\sin(2t)\,\text{V}$ alone and open circuiting the current source $i_s = 12\cos(6t + 10°)\,\text{A}$.

Consider the source is, $v_s = 50\sin(2t)$ V

Convert the source into cosine form.

$$\begin{aligned} v_s &= 50\sin(2t)\ \text{V} \\ &= 50\cos(2t - 90°)\ \text{V} \end{aligned}$$

Determine the capacitive reactance.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)(50\times 10^{-3})} \\ &= -j10\ \Omega \end{aligned}$$

Re-draw the circuit diagram.

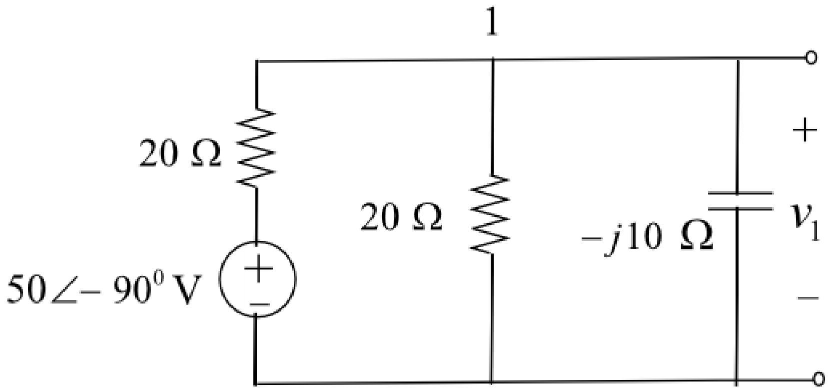


Figure 1

Step 3 of 6

Apply Kirchhoff's current law at node 1.

$$\frac{(50\angle -90°) - v_1}{20} = \frac{v_1}{20} + \frac{v_1}{-j10}$$

$$(50\angle -90°) - v_1 = v_1 + j2v_1$$

$$v_1(2 + j2) = 50\angle -90°$$

$$\begin{aligned} v_1 &= \frac{50\angle -90°}{2 + j2} \\ &= \frac{50\angle -90°}{2.83\angle 45°} \\ &= 17.67\angle -135°\ \text{V} \end{aligned}$$

Convert the equation to time-domain form.

$v_1 = 17.67\cos(2t - 135°)\,\text{V}$ ...... (2)

Step 4 of 6

To find v_2, consider the current source $i_s = 12\cos(6t + 10°)\,\text{A}$ alone and short circuiting the voltage source $v_s = 50\sin(2t)\,\text{V}$.

Determine the capacitive reactance.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(6)(50\times10^{-3})} \\ &= -j3.33\ \Omega \end{aligned}$$

Re-draw the circuit diagram.

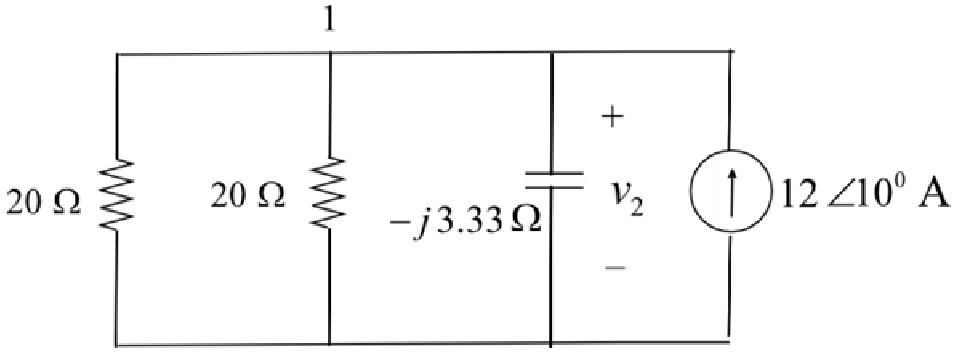


Figure 2

Step 5 of 6

Apply Kirchhoff's current law at node 1.

$$12\angle10° = \frac{v_2}{-j3.33} + \frac{v_2}{20} + \frac{v_2}{20}$$

$$12\angle10° = v_2(0.1 + j0.3)$$

$$\begin{aligned} v_2 &= \frac{12\angle10°}{0.1 + j0.3} \\ &= \frac{12\angle10°}{0.316\angle71.57°} \end{aligned}$$

$$v_2 = 37.95\angle-61.57°\,\text{V}$$

Convert the equation to time-domain form.

$$v_2 = 37.95\cos(6t - 61.57°)\,\text{V} \quad \ldots\ldots (3)$$

Step 6 of 6

Substitute equation (2) and equation (3) in equation (1).

$$v_x = 17.67\cos(2t - 135°) + 37.95\cos(6t - 61.57°)\ \text{V}$$

Therefore, the value of voltage v_x is,

$$\boxed{17.67\cos(2t - 135°) + 37.95\cos(6t - 61.57°)\ \text{V}}.$$